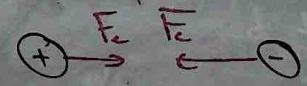
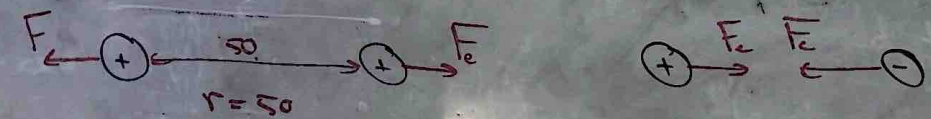


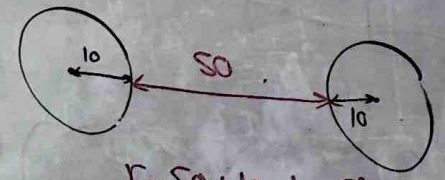
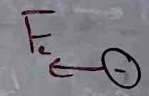
Coulomb's force $\rightarrow$ electrostatic force $\rightarrow$ charges are stationary

electrostatic force can be ① repulsive or ② attractive

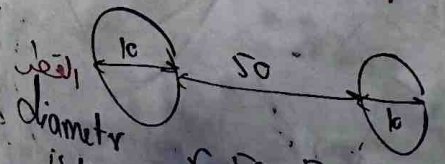
Point charges:



2 spheres of radius 10 cm



$$r = 50 + 10 + 10 = 70$$



$$r = 50 + 5 + 5 = 60$$

Coulomb's force $\rightarrow$ electrostatic force $\rightarrow$ charges are stationary

electrostatic force can be ① repulsive or ② attractive

$$F_e = K_e \frac{|q_1| |q_2|}{r_{1,2}^2}$$

inverse square law

- ① Two charges separated by 2m.
if a separation is 10m.

$$\frac{10}{2} = 5 \rightarrow \frac{F_e}{5^2} = \frac{F_e}{25}$$

- ② Two charge separated by 16m, if they are separated by 4m. Then the force is

$$\frac{16}{4} = 4 \rightarrow 4^2 F_e = 16 F_e$$

- ③ Two charges are doubled and the distance is halved the force

$$F_e = \frac{(2 \times 2) \times 2}{2^2} = 16 F$$

- ④ charges are doubled distance doubled: Force?

$$\frac{2 \times 2}{4} = F$$

The electrostatic force is directly proportional to the product of the two charges and inversely proportional to the square of the distance between them.

$$K_e = \frac{1}{4\pi\epsilon_0} \rightarrow K_e = 9 \times 10^9 \frac{\text{N} \cdot \text{m}^2}{\text{C}^2}$$

in air

$$F_e = K_e \frac{|q_1| |q_2|}{r_{1,2}^2}$$

inverse square law

- ① Two charges separated by 2m.
if a separation is 10m.

$$\frac{10}{2} = 5 \rightarrow \frac{F_e}{5^2} = F_e / 25$$

- ② Two charge separated by 16m, if they are separated by 4m. Then the force is

$$\frac{16}{4} = 4 \rightarrow 4^2 F_e = 16 F_e$$

- ③ Two charges are doubled and the distance is halved
the force

$$F_e = \frac{(2 \times 2) \times 2^2}{1^2} = 16 F$$

- ④ charges are doubled, distance doubled. Force?

$$\frac{2 \times 2}{4} = F$$

The electrostatic force is directly proportional to the product of the two masses and inversely proportional to the square of the distance between them.

$$\rightarrow G = 6.67 \times 10^{-11} \frac{\text{N} \cdot \text{m}^2}{\text{kg}^2}$$

$$F_e = G \frac{m_1 m_2}{r_{1,2}^2}$$

inverse square law

① Gravitational force can only be attractive